

Bernoulli-EKF on SE(3): A Compact Multi-Object Tracker for Mobile Manipulation

Part I — Algorithmic Overview

The system tracks per-object Bernoulli-Gaussian beliefs over rigid body poses on SE(3), conditioned on RGB-D detections from a moving camera mounted on a manipulator. Per frame k it ingests the SLAM base-pose $T_{wb,k}$, the camera-to-base extrinsic $T_{bc,k}$, the gripper-to-base $T_{bg,k}$, the gripper finger width, and a list of SAM2 detections $\{(\text{pid}_d, \text{label}_d, \text{mask}_d, \text{score}_d)\}_d$ projected through depth.

1. State. Each track o carries $(\mu_{bo}, \Sigma_{bo}, r_o, \ell_o, \tau_o)$: $\mu_{bo} \in \text{SE}(3)$ is the object pose in the *base* frame, $\Sigma_{bo} \in \mathbb{R}^{6 \times 6}$ is its covariance in the body-frame tangent at μ_{bo} , $r_o \in [0, 1]$ is the existence probability, ℓ_o a label histogram, and τ_o the SAM2 perception id. *Base-frame storage is the canonical choice*: SLAM uncertainty Σ_{wb} never enters the recursion; the world view is the deterministic composition $T_{wo} = T_{wb} \mu_{bo}$.

2. Predict. Two motion models, applied per-track: world-static (default, $u_k = T_{wb,k}^{-1} T_{wb,k-1}$, applied via the adjoint to Σ plus per-phase Q) or rigid-attached (held set, $\delta_{bg,k} = T_{bg,k} T_{bg,k-1}^{-1}$).

3. Detection conditioning. Subpart suppression (`det_dedup.py`) absorbs detections whose 3D voxel sets are mostly contained in another’s; centroid back-projection drops masks without valid depth.

4. Association. Translation-only Mahalanobis with a χ_3^2 gate, $G_{\text{out},t} = 21.108$, $R_{\text{cam}} = (0.02 \text{ m})^2 I$. Hungarian on the gated cost; SAM2-id continuity bonus $\alpha = 4.4$ breaks ties.

5. Update. For matched pairs on the held track, two safety prefilters guard the Joseph step: held-prefilter (centroid-to- T_{we} within 0.25 m) and innovation-clamp (centroid-to- μ_w within 0.20 m). The Joseph step runs on the *translation block only*; the rotation block and cross-cov are left zeroed (a translation-only observation carries no rotation information). This keeps Σ_{bo} PSD across long runs.

6. Birth and prune. A confirm-3 / score-0.20 / border / near-live (default 5 cm; 25 cm against T_{we} for the held oid) chain controls births. Prune when $r < 10^{-3}$, except hold the held oid above $r = 0.5$.

7. Self-merge. Same-label tracks within 5 cm fuse via information sum, EXCEPT pairs in the relation-protected set (scene graph asserts they are distinct). The held seed is always the keeper.

8. Relations. A VLM/REST backend (`relation_orchestrator.py`) is fired at first call, release transitions, new-oid arrivals, and every 90 frames. Its parent matrix is converted to `RelationEdge` records with the convention **parent = supported, child = supporter** (LLM’s i is the supporter; we emit (j, i, on)). A per-edge EMA ($\alpha = 0.3$, threshold 0.5) smooths flicker.

9. Held-set expansion. The set carried by the gripper is the closure of {held seed} under in/on edges; after each self-merge, EMA keys are remapped (drop $\rightarrow$ keep) so the closure stays valid.

Part II — Detailed Elaboration

II.1 State

`pose_update/gaussian_state.py`.

$$\mathcal{S}_o = (\mu_{bo} \in \text{SE}(3), \Sigma_{bo} \in \mathbb{R}_{\geq 0}^{6 \times 6}, r_o \in [0, 1], \ell_o, \tau_o \in \mathbb{Z}, \text{age}, \text{frames_since_obs}).$$

Body-frame tangent at μ_{bo} : a perturbation $\xi \in \mathfrak{se}(3)$ acts as $\mu' = \mu \text{Exp}(\xi)$ (right-multiply). Ordering $\xi = (\xi_v; \xi_\omega)$ — *translation first then rotation* (matches `pose_update/ekf_se3.py`, lines 5–6, 51, 108). The adjoint

$$\text{Ad}(T) = \begin{bmatrix} R & [t]_{\times} R \\ 0 & R \end{bmatrix}$$

moves a tangent vector through $T = (R, t)$.

II.2 Predict

World-static. A world-static object satisfies $T_{wb,k} \mu_{bo,k} = T_{wb,k-1} \mu_{bo,k-1}$, hence $\mu_{bo,k} = u_k \mu_{bo,k-1}$ with $u_k = T_{wb,k}^{-1} T_{wb,k-1}$.

The covariance recursion now branches on whether the track was observed last frame (`pose_update/gaussian_state.py` `predict_static`, with caller-supplied `unobserved_oids`):

$$\Sigma_{bo,k} = \begin{cases} \Phi(\text{Ad}(u_k) \Sigma_{bo,k-1} \text{Ad}(u_k)^\top + Q_{\text{static}}) & \text{observed at } k-1, \\ \Sigma_{bo,k-1} & \text{unobserved at } k-1 \text{ (frozen keyframe)}. \end{cases} \quad (1)$$

The mean update $\mu_{bo,k} = u_k \mu_{bo,k-1}$ runs in *both* branches, deterministically. Q_{static} schedules on `frames_since_obs`: $1\text{e-}6 I_6$ (idle, < 5), $1\text{e-}5 I_6$ (< 50), $1\text{e-}8 I_6$ thereafter (`ekf_se3.py` $\rightarrow$ `process_noise_for_phase`). Φ is the optional covariance-saturation operator (`ekf_se3.py` $\rightarrow$ `saturate_covariance`)

$$\Phi(P) = \min(1, \text{tr } P_{\max} / \text{tr } P) P, \quad (2)$$

inactive when `cfg.P_max` is `None` (the default). Bernoulli predict on existence is

$$r_{k|k-1} = p_s r_{k-1|k-1}, \quad p_s = 1 \text{ default} \quad (3)$$

(`bernoulli.py` $\rightarrow$ `r_predict`).

Why the unobserved branch freezes Σ_{bo} (keyframe mechanism). The Ad-conjugate transport carries a cross-coupling $[t(u_k)]_{\times} P_{rr} [t(u_k)]_{\times}^{\top}$ that pumps rotation uncertainty into the translation block each frame. For tracks with high σ_{yaw} (e.g. a freshly-released bottle at $\sigma_{yaw} = \pi$ from (17)), the per-frame contribution $\Delta x^2 \cdot \pi^2$ compounds catastrophically (~ 1.2 m over 60 frames at apple.drop). The keyframe-mechanism fix exploits the algebraic identity that, telescoping the deterministic mean update over K frames since the last observation at k_f ,

$$\mu_{bo,K} = u_K \cdots u_{k_f+1} \mu_{bo,k_f} = T_{wb,K}^{-1} T_{wb,k_f} \mu_{bo,k_f}. \quad (4)$$

Composing with the current SLAM pose then gives $T_{wo,K} = T_{wb,K} \mu_{bo,K} = T_{wb,k_f} \mu_{bo,k_f} = \text{constant}$ for a static object, exactly as required. The keyframe is not stored explicitly; $\Sigma_{bo,K} = \Sigma_{bo,k_f}$ is preserved by skipping the cov update. World composition ((9) in §II.5) then uses the *current* $\Sigma_{wb,K}$ for the lever-arm contribution, so the world-frame uncertainty grows only via the SLAM marginal — never via per-frame predict drift. Verified to machine precision on apple_drop oid=1 (fso=224 frames): $\|\mu_{bo,K} - T_{wb,K}^{-1} T_{wb,k_f} \mu_{bo,k_f}\| = 2.22 \cdot 10^{-14}$.

Rigid-attached. For tracks in the held set $\mathcal{H}_k$, $\mu_{bo,k} = \delta_{bg,k} \mu_{bo,k-1}$ with $\delta_{bg,k} = T_{bg,k} T_{bg,k-1}^{-1}$. The driver applies a phase-aware Q_{manip} :

- Seed during grasping/releasing: $\text{diag}((0.02)^2 \times 3, (0.10)^2 \times 3)$ — physical grip-slack.
- Seed during holding: $\text{diag}(2.5e-4 \times 3, 1e-3 \times 3)$ — settled grip.
- Transitive members (apples on tray): always the holding row — they don't deform when the gripper closes around the seed.

The held branch still runs the full Ad-conjugate (no keyframe-mechanism shortcut), so the same translation/rotation cross-pumping applies — bounded in practice by $\|t(\delta_{bg,k})\| \ll 1$ but *not* eliminated. Edge cases with $\sigma_{yaw} \rightarrow \pi$ inside the holding window (re-grasps, release-transition) remain a known caveat; cap-rotation mitigations are an open item.

II.3 Detection conditioning

Subpart suppression (det_dedup.py). Voxelise each mask through depth at 2 cm; for ordered pair (a, b) with $|V_b| \leq |V_a|$, if $|V_b \cap V_a|/|V_b| > 0.8$, absorb b into a (label histograms merge). Catches SAM2 mask splits / re-prompts.

Centroid validity. $c_d^{\text{cam}} = \frac{1}{|m_d \cap \text{valid-depth}|} \sum_{p \in m_d} \text{back_project}(p, \text{depth}, K)$. If the mask has no valid-depth pixels, d is dropped (logged in `centroid_dropped`).

II.4 Association

`pose_update/association.py` $\rightarrow$ `hungarian_associate`. For each (o, d) :

$$\begin{aligned} t_{bo}^{\text{meas}} &= T_{bc} [c_d^{\text{cam}}; 1]_{1:3} \\ R_{bo}^t &= R_{bc} R_{cam} R_{bc}^{\top}, \quad R_{cam} = \text{diag}((0.02 \text{ m})^2 \times 3) \\ \nu_{od} &= t_{bo}^{\text{meas}} - \mu_{bo}[1:3, 4], \quad S_{od} = P_{tt,o} + R_{bo}^t \\ d_{t,od}^2 &= \nu_{od}^{\top} S_{od}^{-1} \nu_{od}. \end{aligned}$$

Cost. Two modes, selected by `BernoulliConfig.enforce_label_match`:

$$C_{od} = d_{t,od}^2 - \alpha_{\text{SAM2}} \mathbf{1}[\tau_o = \text{pid}_d, \tau_o \geq 0] + \beta \mathbf{1}[\hat{\ell}_d \notin \mathcal{H}_o] + \gamma \max(0, 1 - \hat{s}_d). \quad (5)$$

Hard label gate (`enforce_label_match=True`, paper baseline): $\beta = \gamma = 0$, and any cell with mismatched label is forced to the sentinel `_INFEASIBLE = 1012` (big-M). **Soft mode** (`enforce_label_match=False`, used by `tests/visualize_ekf_tracking.py`): $\beta = \text{hungarian_label_penalty}$ (default 0.0; visualizer sets 6.0) and $\gamma = \text{hungarian_score_weight}$ (default 0.0; visualizer sets 2.0). $\alpha_{\text{SAM2}} = 4.4$ in either mode (continuity bonus).

Label-purity gate (soft mode only). The membership test $\hat{\ell}_d \in \mathcal{H}_o$ is not pure dict membership: it is the helper `_label_in_history_meaningful(t_history, d_label, min_obs=5, min_share=0.05)` in `association.py`. A label is considered “meaningfully in the history” iff it has ≥ 5 observations *and* $\geq 5\%$ of the total. This rejects stale trace-amount labels (e.g. 3 obs against 53 990 $\rightarrow$ `False`) that would otherwise bypass the β penalty.

Outer gate. The Hungarian solver minimises $\sum C_{o\sigma(o)}$. Two gates apply:

1. Probabilistic chi²: $d_{t,od}^2 > G_{\text{out},t} = 21.108$ (χ_3^2 at 99.97%) $\rightarrow$ cell infeasible.
2. Hard Euclidean residual cap: $\|\nu_{od}\|_2 > \text{cfg.max_residual_m} \rightarrow$ cell infeasible. Default 0.30 m; set `None` to disable. Catches the inflated-covariance pathology where, after a long miss run with $\sigma_t \approx 0.4$ m, a 0.86 m residual gives $d_t^2 \approx 17 < 21.108$ and the chi² gate becomes meaningless.

$$\sigma^* = \arg \min_{\sigma} \sum_o C_{o\sigma(o)}, \quad \text{drop } (o, d) \text{ if } C_{od} \geq \text{INFEASIBLE}. \quad (6)$$

II.5 Update

Held-track prefilters (`visualize_ekf_tracking.py`, `matched-update` branch). For each accepted match (o, d) with o the held seed:

- Held-prefilter: $\|c_d^w - (T_{wb}T_{bg})_{1:3,4}\|_2 \leq 0.25$ m.
- Innovation clamp: $\|c_d^w - (T_{wb}\mu_{bo})_{1:3,4}\|_2 \leq 0.20$ m.

A failure marks the detection consumed (so it doesn’t re-birth) and skips the EKF step; the rigid-attach predict carries the track.

Joseph step (translation-only), `gaussian_state.py` $\rightarrow$ `update_observation_centroid`. Let $K_{tt} = P_{tt} S^{-1}$ (3×3):

$$\begin{aligned} \delta_t^{\text{base}} &= K_{tt} \nu \\ \mu_{bo}[1:3, 4] &\leftarrow \mu_{bo}[1:3, 4] + \delta_t^{\text{base}} \\ P_{tt}^{\text{post}} &= (I - K_{tt}) P_{tt} (I - K_{tt})^\top + K_{tt} R_{bo}^t K_{tt}^\top \\ \Sigma_{bo}^{\text{post}} &= \text{block-diag}(P_{tt}^{\text{post}}, P_{rr}). \end{aligned}$$

The cross-covariance is zeroed: a translation-only observation provides no information about rotation (we set $R_{rot} = \infty$), so the honest posterior is block-diagonal. An earlier version preserved the prior cross-cov, which broke $P_{rr} - P_{rt}^\top P_{tt}^{-1} P_{rt} \succeq 0$ over many updates; the refactored version is PSD by construction, with an `eigh-clip` belt-and-suspenders.

For pairs where ICP succeeded (`icp_pose.py`), an alternative full-SE(3) update via `joseph_update` in `object_belief.py` replaces the centroid path. The lift to base frame is

$$T_{bo}^{\text{meas}} = T_{bc} \hat{T}^{co}, \quad R^{bo} = \text{Ad}(T_{bc}) \hat{R}^{co} \text{Ad}(T_{bc})^\top, \quad (7)$$

followed by IEKF inner iterations and the Joseph form (`ekf_se3.py` $\rightarrow$ `ekf_update`; `object_belief.py` $\rightarrow$ `joseph_update`; `ekf_se3.py` $\rightarrow$ `huber_weight` for the inner soft / outer hard reweighting on d_t^2 , (10)). The EKF likelihood used by (13) is

$$\log \mathcal{L} = -\frac{1}{2}(\nu^\top S^{-1} \nu + \log \det(2\pi S)), \quad (8)$$

computed in log-space to avoid underflow at $\log \mathcal{L} \sim -50$.

World-frame composition. `gaussian_state.py` $\rightarrow$ `collapsed_object_world` returns the canonical world-frame view consumed by every downstream client (relations, slow tier, gravity-predict input, visualizer):

$$\mu_{wo} = T_{wb} \mu_{bo}, \quad \Sigma_{wo} = \text{Ad}(\mu_{bo}^{-1}) \Sigma_{wb} \text{Ad}(\mu_{bo}^{-1})^\top + \Sigma_{bo}. \quad (9)$$

Σ_{wb} is read from the current SLAM marginal at query time, so the world cov tracks any SLAM drift / loop-closure shift even when no new object observation has arrived. Combined with the keyframe-mechanism freeze of Σ_{bo} ((1)), an unobserved static track’s world cov grows *only* via the Σ_{wb} -lever — the user-intended behaviour.

World-frame overwrite (release hook). `gaussian_state.py` $\rightarrow$ `overwrite_object_pose(oid, T_{wo}, P_{wo})` forces the EKF state for one oid from a world-frame (T, P) input. Used by `orchestrator_gaussian.py` $\rightarrow$ `maybe_gravity_predict` (and the RBPF analogue in `orchestrator.py`) at the release transition to install (17)’s landing prior. The body-frame storage becomes $\mu_{bo} = T_{wb}^{-1} T_{wo}$ and $\Sigma_{bo} = P_{wo}$ (the right-trivialised tangent transport reduces to the identity at the new μ_{bo}). After the overwrite, the keyframe-mechanism freeze in (1) keeps the gravity-supplied cov ($\sigma_{yaw} = \pi$, $\sigma_{xy} \sim 5\text{--}10\text{ cm}$) intact across subsequent unobserved frames.

Per-track observation chain (diagnostic). Each track also accumulates an append-only chain $\mathcal{H}^{(i)} = [(k, \hat{T}_k^{co}, \hat{R}_k^{co}), \dots]$ of camera-frame ICP measurements (`pose_update/obs_chain.py`). **The chain is NOT the canonical world-frame source** — that role belongs to `collapsed_object_world` ((9)). The chain is computed in parallel by the visualizer (`tests/visualize_ekf_tracking.py`) for cross-checking and is the natural substrate for a future loop-closure-aware smoother (the slow-tier `pose_update/factor_graph.py` consumes the same structure when promoted).

Huber reweight.

$$w(d^2; G_{\text{in}}, G_{\text{out}}) = \begin{cases} 1 & d^2 \leq G_{\text{in}}, \\ G_{\text{in}}/d^2 & G_{\text{in}} < d^2 \leq G_{\text{out}}, \\ 0 & d^2 > G_{\text{out}}. \end{cases} \quad (10)$$

Defaults: $G_{\text{in},t} = 7.815$ (χ_3^2 at 95%), $G_{\text{out},t} = 21.108$.

II.6 Birth and prune

Birth gates (`orchestrator.py` $\rightarrow$ `birth_admissible` and `birth_gating.py` $\rightarrow$ `is_near_live_track`):

1. Border: bbox not within `birth_border_margin_px=2` px of edge.
2. Confirm- k : tracker-side pending counter $\geq k = \text{birth_confirm_k}=3$ (counts the tracker's unmatched-streak frames, not perception's n_{obs} which resets on SAM2 id reissue).
3. Score floor: rolling-max score $\geq \text{birth_score_min}=0.20$.
4. Near-live: world-frame centroid $> \text{birth_min_dist_m}=0.05$ m from any same-label live track. For the held oid the radius widens to `held.birth_radius_m=0.25` m and T_{we} is checked as a secondary anchor.
5. TTL: pending entries that go `birth_pending_ttl_frames=30` frames without a matching arrival are evicted (one second at 30 Hz; survives short perception dropouts).

Survivors mint a fresh oid via `_next_track_id()`, run ICP once on admission (`icp_pose.py` $\rightarrow$ `PoseEstimate.estimate` with `T_co_init=None`), and seed

$$r_0 = \frac{\lambda_b \hat{s}_d}{\lambda_b \hat{s}_d + \lambda_c} \quad (11)$$

(`bernoulli.py` $\rightarrow$ `r_birth`; defaults $\lambda_b = \lambda_c = 1 \rightarrow r_0 = \hat{s}_d$). Birth covariance defaults to

$$P_{\text{new}} = \hat{R}^{co} \quad (\text{init_cov_from_R=True, default}) \quad (12)$$

and falls back to $\text{diag}((5 \text{ cm})^2 \times 3, (0.05 \text{ rad})^2 \times 3)$ when `init_cov_from_R=False`.

Existence updates. For matched pairs ((13); `bernoulli.py` $\rightarrow$ `r_assoc_update_loglik`)

$$r' = \frac{(1 - p_s) + p_s p_d \mathcal{L} r}{1 - p_s + p_s r}, \quad (13)$$

and for missed tracks ((14); `bernoulli.py` $\rightarrow$ `r_miss_update`; $\tilde{p}_d = p_d p_v$)

$$r' = \frac{(1 - \tilde{p}_d) r}{1 - \tilde{p}_d r}, \quad (14)$$

with p_v from `pose_update/visibility.py` (see (16)).

Prune. $r_o < \text{r_min}=10^{-3}$ deletes, EXCEPT the held seed, whose existence is floored at `r_held_floor=0.5` (survives long held-occlusion).

II.7 Self-merge

`orchestrator.py` $\rightarrow$ `_self_merge_pass`. Inputs: held seed s , protected pairs $\mathcal{P}$ derived from the relation edge set $\mathcal{E}$ emitted by `relation_orchestrator.py` (§II.8) — the parent/child convention is *parent* = *supported*, *child* = *supporter*, so a relation (p, c, on) contributes the unordered pair $(\min(p, c), \max(p, c))$ to $\mathcal{P}$:

$$\mathcal{P} = \{ (\min(p, c), \max(p, c)) : (p, c, \cdot) \in \mathcal{E} \}.$$

Same-label pairs in $\mathcal{P}$ are protected from collapse — the scene graph asserts they are physically distinct (apple-on-tray, etc.). Information-sum fusion (with the keep mean as linearisation point) is

$$\begin{aligned} P_{\text{new}}^{-1} &= P_{\text{keep}}^{-1} + P_{\text{drop}}^{-1}, \\ \mu_{\text{new}} &= \mu_{\text{keep}} \text{Exp}(P_{\text{new}} P_{\text{drop}}^{-1} \text{Log}(\mu_{\text{keep}}^{-1} \mu_{\text{drop}})). \end{aligned} \quad (15)$$

Algorithm 1 Self-merge pass.

```

1: for each pair  $(o_i, o_j)$  with  $\ell_i = \ell_j$  do
2:   if  $(\min(o_i, o_j), \max(o_i, o_j)) \in \mathcal{P}$  then continue
3:   end if
4:    $d \leftarrow \|\mu_w(o_i) - \mu_w(o_j)\|_2$ 
5:   if  $d > 0.05$  then continue
6:   end if
7:   Push  $(d, d_t^2, o_i, o_j)$  as candidate.
8: end for
9: Sort candidates by  $d$  ascending.
10: for each  $(d, d_t^2, o_i, o_j)$  do
11:   if  $o_i$  or  $o_j$  already absorbed then continue
12:   end if
13:   Choose keep  $\leftarrow s$  if  $s \in \{o_i, o_j\}$ , else higher- $r$ , else lower oid.
14:   Information-sum-fuse  $\mu, \Sigma$ ; merge label histograms / sam2_tau / T_oe / r.
15:   Append  $\{\text{keep}, d, d_t^2\}$  to merges.
16: end for

```

The protected-pairs gate is what stops an apple-on-tray pair from collapsing once their world centroids drift within 5 cm under occlusion bias.

II.8 Visibility

`pose_update/visibility.py` $\rightarrow$ `visibility_p_v`. For each track i , sample object-surface points $\{x_j^{(i)}\}_{j=1}^M$ (paper specifies projecting the TSDF shape summary $\mathcal{G}^{(i)}$; current implementation uses a Fibonacci sphere of radius r_{obj} around μ_{bo}). Lift each through T_{bc} into camera, depth-test against the current depth image:

$$p_v^{(i)} = \frac{\#\{j : x_j^{(i)} \text{ in-frustum and passes depth test}\}}{\#\{j : x_j^{(i)} \text{ in-frustum}\}} \in [0, 1]. \quad (16)$$

$p_v = 1$ when no in-frustum sample is occluded (fully visible / no evidence of occlusion); $p_v = 0$ when fully out-of-FOV or fully occluded. The continuous probability composes naturally with (14); the four-state machine (present / occluded / absent / not-in-view) collapses into a single weight on the miss-frame likelihood.

II.9 Relations

Trigger gate (`relation_utils.py` $\rightarrow$ `should_recompute_relations`): fires at first call, release transitions, new-oid arrivals, or every `relation_every_n_frames=90` frames. The grasp-transition trigger was removed (LLM at grasp onset has incomplete information).

Backend (`relation_orchestrator.py` + `relation_client.py`). On a triggered frame, the orchestrator builds normalised bboxes per oid (from `det_to_oid`) and calls `LLMRelationClient` or `RESTRelationClient`. The backend returns $p \in [0, 1]^{N \times N}$ with $p_{ij} = “j \text{ rests on } i”$. Disk cache (`CachedRelationClient`) keyed by (frame, N , rounded bboxes).

Convention swap. Held-set expansion uses `parent=supported`, `child=supporter`. The LLM’s (i, j) encodes the opposite (j rests on i). Edge emission swaps:

$$\text{edge} = \text{RelationEdge}(\text{parent} = j, \text{child} = i, \text{type} = \text{on}, \text{score} = p_{ij}).$$

A regression here was responsible for held-set expansion failing — the closure walks edges with `child` $\in S$, so a backward emission means no propagation.

EMA (`orchestrator.py` $\rightarrow$ `RelationFilter`). For each key $k = (\text{parent}, \text{child}, \text{type})$:

$$\text{ema}_k(t) = \alpha \text{raw}_k(t) + (1 - \alpha) \text{ema}_k(t - 1), \quad \alpha = 0.3.$$

Edges emitted when $\text{ema} \geq 0.5$; pruned when $\text{ema} \leq 0.01$. Half-life ~ 2 trigger calls ~ 180 frames.

Remap after merges. When self-merge produces $\{(\text{drop}_i, \text{keep}_i)\}$, every EMA key and emitted edge has `parent/child` rewritten via the closure of `drop_to_keep`. Self-loops drop, duplicate keys merge by max.

II.10 Held-set expansion

`relation_utils.py` $\rightarrow$ `expand_held_with_relations`.

```

1:  $S \leftarrow \{s\}$ 
2: for iter = 1...8 do
3:   changed  $\leftarrow$  false
4:   for each edge  $(p, c, t) \in \mathcal{E}$ ,  $t \in \{\text{in}, \text{on}\}$  do
5:     if  $c \in S$  and  $p \notin S$  then  $S \leftarrow S \cup \{p\}$ ; changed $\leftarrow$ true
6:   end if
7: end for
8: if not changed then break
9: end if
10: end for
11: return  $S$ 

```

The caller post-filters S against currently-live oids to drop references to pruned tracks the EMA hasn’t yet decayed.

II.11 Gravity-aware predict at release

When the gripper releases an object, the rigid-attachment predict no longer applies and the static predict over-states the prior (the object physically falls). This subsystem replaces the EKF mean and covariance for the just-released oid with the post-settling prediction in a single step at the `RELEASING` $\rightarrow$ `IDLE` transition.

Voxel observability. `pose_update/voxel_observability.py` maintains a dense 3-state grid (`UNSEEN` / `EMPTY` / `OCCUPIED`) plus per-voxel `n_hit` + `n_pass` counters with hysteresis ($\geq \text{n_min_hit}=2$ confirming hits and $\geq \text{n_min_pass}=3$ pass-through rays before transition). Updated each frame from depth: each pixel’s ray marks voxels traversed before the depth hit as pass-pending and the voxel at the hit as hit-pending. Default workspace 5 m $\times$ 5 m $\times$ 3 m at 5 cm voxel size.

Object dynamics property. `pose_update/object_dynamics.py` hosts a per-label table mapping each detection label to $(e, \mu, \text{shape}, r_{\text{obj}})$. Restitution defaults from agricultural / engineering literature: apple $e = 0.30$, milkbox 0.40, cola 0.50, cup 0.40, pot 0.40, flowerpot 0.30. Unknown labels fall back to $e = 0.40$, $r = 0.05$ (DEFAULT_DYNAMICS). A perception-side override hook lets a learned estimator (e.g. Galileo) supersede the table per-detection.

Parametric dispersion model (`pose_update/gravity_predict.py` $\rightarrow$ `predict_landing_pose`). For drop height h (resolved via voxel-driven raycast, below):

$$\begin{aligned}
v_{\text{impact}} &= \sqrt{2gh}, & t_{\text{fall}} &= \sqrt{2h/g}, \\
\sigma_{\text{bounce}} &= \varepsilon_{\text{rough}} v_{\text{impact}} / \max(0.1, 1 - e), \\
\sigma_{\text{shape}} &= r_{\text{obj}} \text{footprint_factor}(\text{shape}), \\
\sigma_{\text{rel-v}} &= \frac{1}{2} \|\mathbf{v}_{\text{rel,xy}}\| t_{\text{fall}}, \\
\sigma_{xy}^2 &= \sigma_{\text{bounce}}^2 + \sigma_{\text{shape}}^2 + \sigma_{\text{rel-v}}^2, \\
\sigma_z &= \frac{1}{2} r_{\text{obj}} (1 - e/2), \\
\sigma_{\text{yaw}} &= \pi (1 - \exp(-h(1 - e^2)/r_{\text{obj}})).
\end{aligned} \tag{17}$$

Default $\varepsilon_{\text{rough}} = 5 \cdot 10^{-3} \text{ m} / (\text{m s}^{-1})$; **shape_footprint_factor**: spherical 0.25, cylindrical 0.50, box 0.70, irregular 1.00. Higher restitution increases σ_{bounce} — physically, a rubber ball bounces several times each accruing scatter, while a clay ball lands once and stays.

Voxel-driven h branch. The downward raycast (`VoxelObservability.raycast_down`) classifies the column into one of four states; each adjusts h , landing z , and the σ_z floor:

- **hit_occupied**: clean surface at h^* ; $h = h^*$, σ_z from (17).
- **mixed_unseen**: surface known but UNSEEN voxels above it; $h = h^*$, σ_z extra-inflated by $(h_u - h^*)/2$ to reflect path uncertainty.
- **all_unseen**: surface unknown; $h = (h_u + h_{\text{floor}})/2$ (*midpoint*, not worst case), $\sigma_z = (h_u - h_{\text{floor}})/2$.
- **all_empty**: column observed empty all the way to the workspace floor; $h = h_{\text{floor}}$, $\sigma_z = r_{\text{obj}}$.

The midpoint choice for **all_unseen** gives the EKF a useful prior that subsequent observations can correct quickly, instead of pushing the mean to an extreme that's unlikely to be right.

Trigger. Three production sites all detect the FSM exiting {HOLDING, RELEASING} and call into (17):

- `pose_update/orchestrator.py` $\rightarrow$ `_maybe_gravity_predict` (RBPF backend) — replaces every particle's (T_{wo}, P_{wo}) via `RBPFState.override_object_pose`.
- `pose_update/orchestrator_gaussian.py` $\rightarrow$ `_maybe_gravity_predict` (Gaussian backend) — calls `GaussianState.override_object_pose` (see §II.5 world-frame overwrite). Wired in `TwoTierOrchestratorGaussian.step` immediately after the fast tier.
- `tests/visualize_ekf_tracking.py` (instrumented tracker harness) — mutates `state.objects[oid].{mu_bo, cov_bo}` inline.

Subsequent frames see the released track in `unobserved_oids`, so (1)'s keyframe freeze preserves the gravity prior (in particular $\sigma_{\text{yaw}} = \pi$ does *not* pump σ_{xy} via cross-coupling).

Toggle. `BernoulliConfig.gravity_predict` (default `True`) gates the RBPF hook; `TwoTierOrchestratorGaussian` (constructor flag, default `True`) gates the Gaussian hook. `BernoulliConfig.workspace_floor_z` / `TwoTierOrchestratorGaussian.workspace_floor_z` (default `-1.0` m) sets the column terminator. With `voxel_obs` unset, both hooks short-circuit and the legacy static-Q-only release behaviour is preserved exactly.

II.12 Gripper phase FSM

`pose.update/gripper_state.py` $\rightarrow$ `GripperPhaseTracker`. Four phases: `IDLE` $\rightarrow$ `GRASPING` $\rightarrow$ `HOLDING` $\rightarrow$ `RELEASING` $\rightarrow$ `IDLE`. **Project assumption:** the gripper begins fully open; ANY closing of the fingers registers as a grasp action. The state machine therefore detects “closed” as a drop $\geq \text{close_delta_m}=0.005$ m below the rolling open baseline (5 mm captures any non-trivial finger motion while ignoring proprioception jitter), in addition to the absolute thresholds. Driven by:

- Width threshold crossings with hysteresis (`closed_width_m=0.025` m, `open_width_m=0.040` m, `close_delta_m=0.005` m below baseline).
- Stability test (last `history_size=5` widths span $< \text{motion_threshold_m}=0.01$ m).
- Min-frames timer (`min_transition_frames=5`) holds the transition phases through brief instability.

At the `IDLE` $\rightarrow$ `GRASPING` edge, the held oid is selected by `GraspOwnerDetector`: rank detections by inside- gripper-volume containment fraction ($= \text{count_inside} / \text{total_pts}$); tiebreak by smallest mask area. Falls back to nearest-live-track within 0.30 m of T_{we} . Self-merges propagate via `apply_merges`: `drop = held_obj_id` $\rightarrow$ switch to the keep.

II.13 Frame-by-frame algorithm

Algorithm 2 One frame of the Bernoulli-EKF tracker (driver loop).

```

1: Inputs:  $T_{wb}, T_{bc}, T_{bg}, w$ , RGB, depth, detections.
2:  $g \leftarrow \text{GRIPPERPHASETRACKER.STEP}(w, \dots)$ 
3:  $\text{RELATION\_PIPELINE.MAYBE\_UPDATE}(\dots)$ 
4:  $\mathcal{H} \leftarrow \text{EXPAND}(g.\text{held}, \mathcal{E}_t) \cap \text{live oids}$ 
5:  $\mathcal{U} \leftarrow \{o : \text{frames\_since\_obs}[o] > 0\}$  ▷ keyframe-frozen set
6:  $\text{PREDICT\_STATIC}(\text{skip} = \mathcal{H}, \text{unobserved} = \mathcal{U}); \text{RIGID\_ATTACHMENT\_PREDICT}(\text{each } o \in \mathcal{H})$ 
7:  $\text{SUPPRESS\_SUBPART\_DETECTIONS}; \text{CENTROID\_CAM\_FROM\_MASK}; \text{gather } \mathbf{dets\_for\_assoc}$ 
8:  $\text{HUNGARIAN\_ASSOCIATE} \rightarrow \text{matches, unmatched\_tracks, unmatched\_dets}$ 
9: for each match  $(o, d)$  do
10:   if  $o = g.\text{held}$  and (held-prefilter or innov-clamp triggers) then skip update
11:   end if
12:    $\text{UPDATE\_OBSERVATION\_CENTROID}; \text{update } r_o \text{ via match log-likelihood}$ 
13: end for
14: for each unmatched track  $o$  do
15:   Update  $r_o$  via miss equation; floor at  $\mathbf{r\_held\_floor}$  if  $o = g.\text{held}$ 
16: end for
17: for each unmatched detection  $d$  do
18:   if  $\text{BIRTH\_ADMISSIBLE}(d)$  and not  $\text{IS\_NEAR\_LIVE\_TRACK}(d)$  then
19:     Mint new oid, run ICP once, set  $r_0 = 0.4$ 
20:   end if
21: end for
22: Prune oids with  $r_o < r_{\min}$  (skip held seed)
23:  $\mathcal{P} \leftarrow \{(\min(p, c), \max(p, c)) : (p, c, \cdot) \in \mathcal{E}_t\}$ 
24:  $\text{\_SELF\_MERGE\_PASS}(\text{held} = g.\text{held}, \text{protected} = \mathcal{P})$ 
25:  $g.\text{APPLY\_MERGES}(\text{merges}); \text{RELATION\_PIPELINE.REMAP\_AFTER\_MERGES}(\text{merges})$ 
26:  $\text{\_MAYBE\_GRAVITY\_PREDICT}(g)$  ▷ release transition: overwrite the just-released oid via (17)

```

II.14 Numerical defaults summary

R_{cam}	$\text{diag}((0.02\text{ m})^2)$	centroid measurement
Q_{static}	$1e-6 \rightarrow 1e-5 \rightarrow 1e-8$	schedule on <code>frames_since_obs</code>
$Q_{manip\text{ seed}} \text{ (graspg./relsg.)}$	$\text{diag}((0.02)^2 \times 3, (0.10)^2 \times 3)$	per frame
$Q_{manip\text{ seed}} \text{ (holdg.)}$	$\text{diag}(2.5e-4 \times 3, 1e-3 \times 3)$	per frame
$Q_{manip\text{ transit}}$	holding row above	apples-on-tray
$G_{out,t}$	21.108	χ^2_3 at 99.97%
$G_{in,t}$	7.815	χ^2_3 at 95% Huber inner
α_{SAM2}	4.4	continuity bonus
<code>max_residual_m</code>	0.30 m	hard Euclidean residual cap (None disables)
<code>P_max</code>	None	covariance saturation cap (no cap default)
<code>P_min_diag</code>	None	per-axis P floor (None = no floor)
<code>birth_min_dist_m</code>	0.05	default near-live radius
<code>birth_border_margin_px</code>	2	image-edge margin
<code>birth_pending_ttl_frames</code>	30	pending-birth TTL ($\sim 1\text{ s}$ @ 30 Hz)
<code>held_birth_radius_m</code>	0.25	wider for held oid
<code>held_meas_radius_m</code>	0.25	T_{we} prefilter (driver-side)
<code>held_meas_innov_max_m</code>	0.20	innov clamp (driver-side)
<code>r_min</code>	10^{-3}	prune threshold
<code>r_held_floor</code>	0.50	held existence floor
<code>birth_confirm_k</code>	3	tracker-side unmatched-streak count
<code>birth_score_min</code>	0.20	rolling-max score floor
<code>self_merge_trans_m</code>	0.05	Euclidean merge gate
<code>dedup_voxel_size_m</code>	0.02	subpart suppression voxel size
<code>dedup_containment_thresh</code>	0.8	subpart suppression threshold
<code>hungarian_label_penalty</code> (β)	0.0 default	soft mode; visualizer sets 6.0
<code>hungarian_score_weight</code> (γ)	0.0 default	soft mode; visualizer sets 2.0
<code>closed_width_m, open_width_m</code>	0.025, 0.040	FSM hysteresis
<code>close_delta_m</code>	0.005	drop-below-baseline = grasp
<code>grasp_radius_m</code>	0.30	nearest-track gate at grasp onset
EMA α , threshold	0.3, 0.5	relation smoothing
<code>relation_every_n_frames</code>	90	periodic LLM tick
<code>gravity_predict</code>	True	gate one-shot release-time gravity hook
<code>workspace_floor_z</code>	-1.0 m	column terminator for <code>raycast_down</code>
ε_{rough}	$5 \cdot 10^{-3}$	bounce-roughness coefficient ($\text{m}/\text{m s}^{-1}$)
voxel grid (default)	5 cm at $5 \times 5 \times 3\text{ m}$	<code>voxel_observability.py</code>
<code>n_min_hit, n_min_pass</code>	2, 3	voxel hysteresis thresholds

II.15 Implementation index

- State / predict: `pose_update/gaussian_state.py`, `pose_update/object_belief.py`, `pose_update/ekf_se3`
- Detection conditioning: `pose_update/det_dedup.py`, `pose_update/icp_pose.py`
- Association / update: `pose_update/association.py`, `pose_update/orchestrator.py`, `gaussian_state.py`
- Birth / prune: `pose_update/orchestrator.py::birth_admissible`, `pose_update/birth_gating.py`
- Self-merge: `pose_update/orchestrator.py::_self_merge_pass`

- Relations: `pose_update/relation_orchestrator.py`, `pose_update/relation_client.py`, `pose_update/relation_utils.py`
- Gripper FSM: `pose_update/gripper_state.py`, `pose_update/grasp_owner_detector.py`
- Visibility: `pose_update/visibility.py`
- Gravity-aware release predict: `pose_update/voxel_observability.py`, `pose_update/object_dynamics.py`, `pose_update/gravity_predict.py`
- Driver / harness: `tests/visualize_ekf_tracking.py`